Pre Mid-Sem Test

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The survey dataset from the MASS package contains anthropometric and lifestyle information of students. In this notebook, descriptive statistical techniques are applied to explore body mass index (BMI), exercise patterns, and physiological measures such as pulse rate. The goal is to understand central tendency, variability, and relationships between variables using graphical and numerical summaries.

```r
##############################################################
# STEP 1: Load Required Package and Dataset
##############################################################

# The MASS package contains the 'survey' dataset
# This dataset includes anthropometric and lifestyle
# information of university students.

library(MASS)

# Load dataset
data(survey)

# Inspect structure to understand variable types
# This helps us decide which variables are numeric or categorical
str(survey)
```

```
## 'data.frame':    237 obs. of  12 variables:
##  $ Sex   : Factor w/ 2 levels "Female","Male": 1 2 2 2 2 1 2 1 2 2 ...
##  $ Wr.Hnd: num  18.5 19.5 18 18.8 20 18 17.7 17 20 18.5 ...
##  $ NW.Hnd: num  18 20.5 13.3 18.9 20 17.7 17.7 17.3 19.5 18.5 ...
##  $ W.Hnd : Factor w/ 2 levels "Left","Right": 2 1 2 2 2 2 2 2 2 2 ...
##  $ Fold  : Factor w/ 3 levels "L on R","Neither",..: 3 3 1 3 2 1 1 3 3 3 ...
##  $ Pulse : int  92 104 87 NA 35 64 83 74 72 90 ...
##  $ Clap  : Factor w/ 3 levels "Left","Neither",..: 1 1 2 2 3 3 3 3 3 3 ...
##  $ Exer  : Factor w/ 3 levels "Freq","None",..: 3 2 2 2 3 3 1 1 3 3 ...
##  $ Smoke : Factor w/ 4 levels "Heavy","Never",..: 2 4 3 2 2 2 2 2 2 2 ...
##  $ Height: num  173 178 NA 160 165 ...
##  $ M.I   : Factor w/ 2 levels "Imperial","Metric": 2 1 NA 2 2 1 1 2 2 2 ...
##  $ Age   : num  18.2 17.6 16.9 20.3 23.7 ...
```

```r
# View first few observations
head(survey)
```

```
##      Sex Wr.Hnd NW.Hnd W.Hnd    Fold Pulse    Clap Exer Smoke Height      M.I
## 1 Female   18.5   18.0 Right  R on L    92    Left Some Never 173.00   Metric
## 2   Male   19.5   20.5  Left  R on L   104    Left None Regul 177.80 Imperial
## 3   Male   18.0   13.3 Right  L on R    87 Neither None Occas     NA     <NA>
```

```
## 4   Male   18.8   18.9 Right  R on L    NA Neither None Never 160.00   Metric
## 5   Male   20.0   20.0 Right Neither    35   Right Some Never 165.00   Metric
## 6 Female   18.0   17.7 Right  L on R    64   Right Some Never 172.72 Imperial
##      Age
## 1 18.250
## 2 17.583
## 3 16.917
## 4 20.333
## 5 23.667
## 6 21.000
```

```r
##############################################################
# STEP 2: Select Variables and Clean Data
##############################################################

# Select relevant variables
selected_data <- survey[, c("Sex", "Height", "Pulse",
                            "Exer", "Smoke", "Age", "Wr.Hnd")]

# Remove missing values to avoid bias in summary statistics
clean_data <- na.omit(selected_data)

# Check cleaned structure
str(clean_data)
```

```
## 'data.frame':    169 obs. of  7 variables:
##  $ Sex   : Factor w/ 2 levels "Female","Male": 1 2 2 1 2 1 2 2 1 1 ...
##  $ Height: num  173 178 165 173 183 ...
##  $ Pulse : int  92 104 35 64 83 74 72 90 80 66 ...
##  $ Exer  : Factor w/ 3 levels "Freq","None",..: 3 2 3 3 1 1 3 3 1 3 ...
##  $ Smoke : Factor w/ 4 levels "Heavy","Never",..: 2 4 2 2 2 2 2 2 2 2 ...
##  $ Age   : num  18.2 17.6 23.7 21 18.8 ...
##  $ Wr.Hnd: num  18.5 19.5 20 18 17.7 17 20 18.5 17 19.5 ...
##  - attr(*, "na.action")= 'omit' Named int [1:68] 3 4 12 13 15 16 19 25 26 29 ...
##   ..- attr(*, "names")= chr [1:68] "3" "4" "12" "13" ...
```

```r
# Check how many observations remain
nrow(clean_data)
```

```
## [1] 169
```

```r
##############################################################
# STEP 3: Descriptive Statistics for Height
##############################################################

# Mean:
# The arithmetic average height of all individuals.
# It represents the central location of the distribution.
mean_height   <- mean(clean_data$Height)

# Median:
# The middle value when heights are arranged in order.
```

```r
# It is resistant to extreme values and helps assess skewness.
median_height <- median(clean_data$Height)

#Standard Deviation (SD):
# Measures the average deviation of individual heights from the mean height.
# Larger SD indicates greater variability.
sd_height     <- sd(clean_data$Height)

# Variance:
# Variance is the square of the standard deviation.
# It measures overall spread of the data in squared units.
var_height    <- var(clean_data$Height)

# Interquartile Range (IQR):
# IQR = Q3 - Q1.
# It measures the spread of the middle 50% of observations.
# Unlike SD, it is robust to extreme values (outliers).
# It helps understand variability without being influenced by extremes.
IQR_height    <- IQR(clean_data$Height)

# Coefficient of Variation
# CV expresses variability relative to the mean.
# It is calculated as (SD / Mean) * 100.
# Useful for comparing variability across variables with different units.
cv_height <- (sd_height / mean_height) * 100

# Create a summary table
height_summary <- data.frame(
  Statistic = c("Mean", "Median", "Standard Deviation",
                "Variance", "Interquartile Range",
                "Coefficient of Variation (%)"),
  Value = c(mean_height, median_height, sd_height,
            var_height, IQR_height, cv_height)
)

# Display table
height_summary
```

```
##                      Statistic      Value
## 1                         Mean 172.522840
## 2                       Median 171.000000
## 3           Standard Deviation   9.948264
## 4                     Variance  98.967954
## 5          Interquartile Range  15.000000
## 6 Coefficient of Variation (%)   5.766346
```

```r
mean_age   <- mean(clean_data$Age)
sd_age     <- sd(clean_data$Age)

mean_wr    <- mean(clean_data$Wr.Hnd)
sd_wr      <- sd(clean_data$Wr.Hnd)

mean_age
```

```
## [1] 20.41573
```

```r
sd_age
```

```
## [1] 6.085532
```

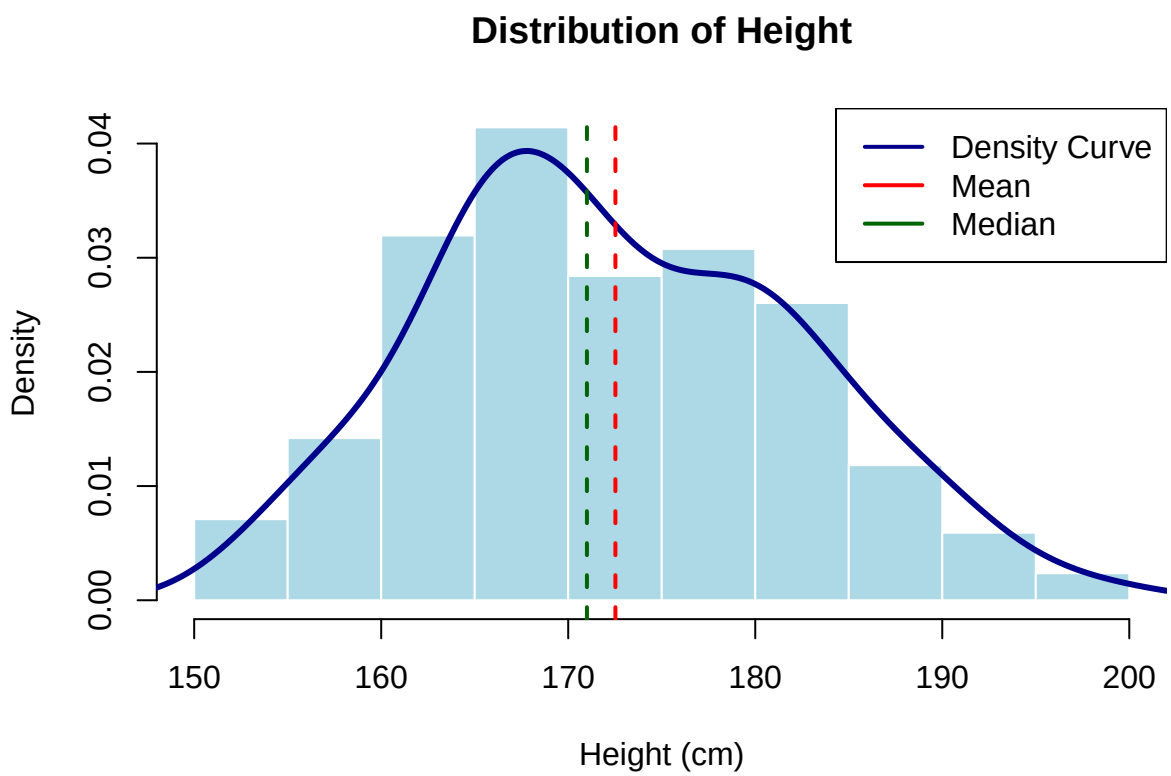
```r
mean_wr
```

```
## [1] 18.76805
```

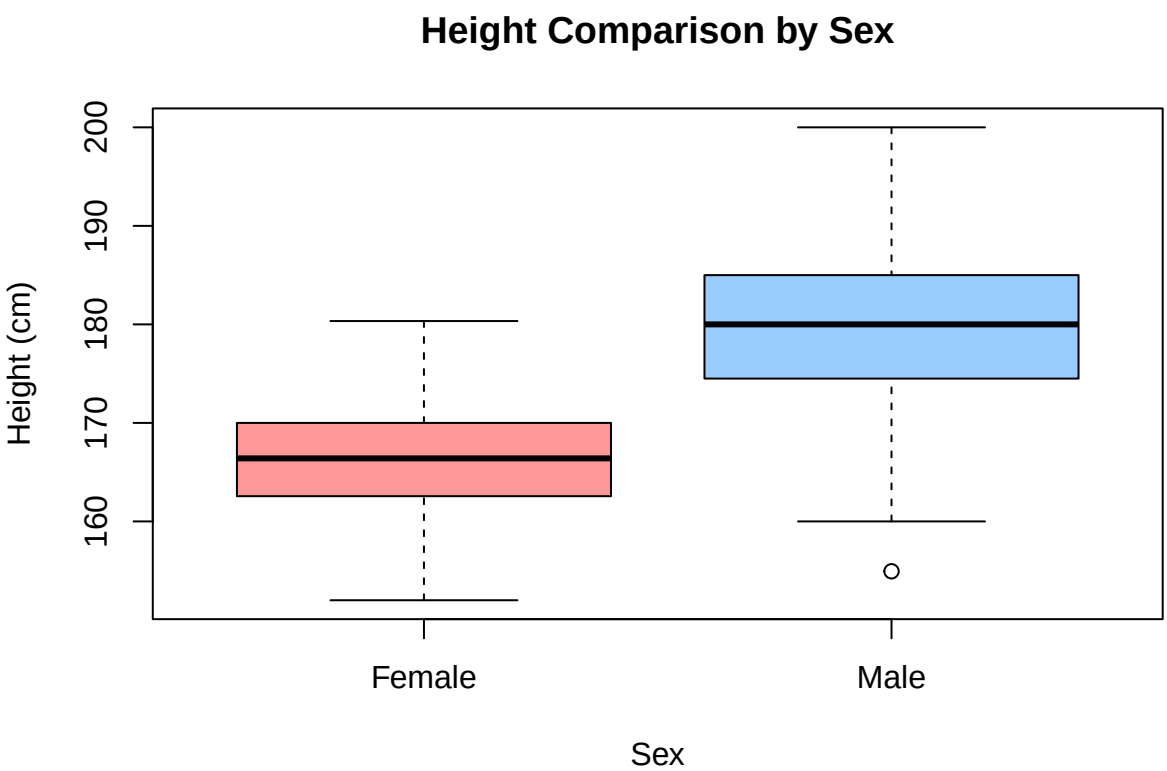
```r
sd_wr
```

```
## [1] 1.964851
```

```r
##############################################################
# STEP 4: Histogram of Height with Density Overlay
##############################################################

# A histogram shows the distribution shape (symmetry/skewness).
# Density curve smooths the distribution.
# Mean and median lines help assess skewness visually.

hist(clean_data$Height,
     probability = TRUE,
     col = "lightblue",
     border = "white",
     main = "Distribution of Height",
     xlab = "Height (cm)")

lines(density(clean_data$Height),
      col = "darkblue",
      lwd = 3)

abline(v = mean_height,
       col = "red",
       lwd = 2,
       lty = 2)

abline(v = median_height,
       col = "darkgreen",
       lwd = 2,
       lty = 2)

legend("topright",
       legend = c("Density Curve", "Mean", "Median"),
       col = c("darkblue", "red", "darkgreen"),
       lwd = 2)
```




```
##############################################################
# STEP 5: Boxplot of Height by Sex
##############################################################

# Boxplots allow comparison of median, spread (IQR),
# and outliers between males and females.

boxplot(Height ~ Sex,
        data = clean_data,
        col = c("#FF9999", "#99CCFF"),
        notch = FALSE,
        main = "Height Comparison by Sex",
        ylab = "Height (cm)")
```

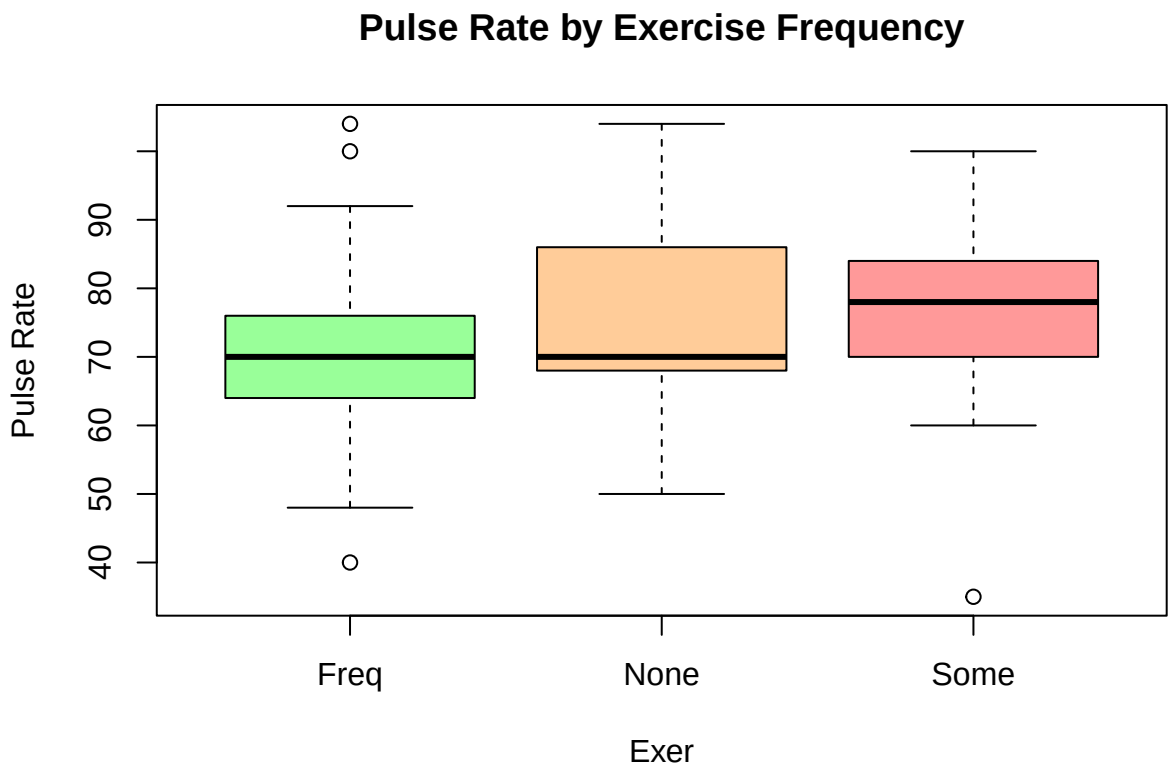



```
##############################################################
# STEP 6: Descriptive Statistics for Pulse
##############################################################

# (Same statistical measures as calculated for Height)
mean_pulse   <- mean(clean_data$Pulse)
median_pulse <- median(clean_data$Pulse)
sd_pulse     <- sd(clean_data$Pulse)
IQR_pulse    <- IQR(clean_data$Pulse)

pulse_stats <- data.frame(
  Statistic = c("Mean", "Median", "Standard Deviation", "Interquartile Range"),
  Value = round(c(mean_pulse, median_pulse, sd_pulse, IQR_pulse), 2)
)

pulse_stats
```

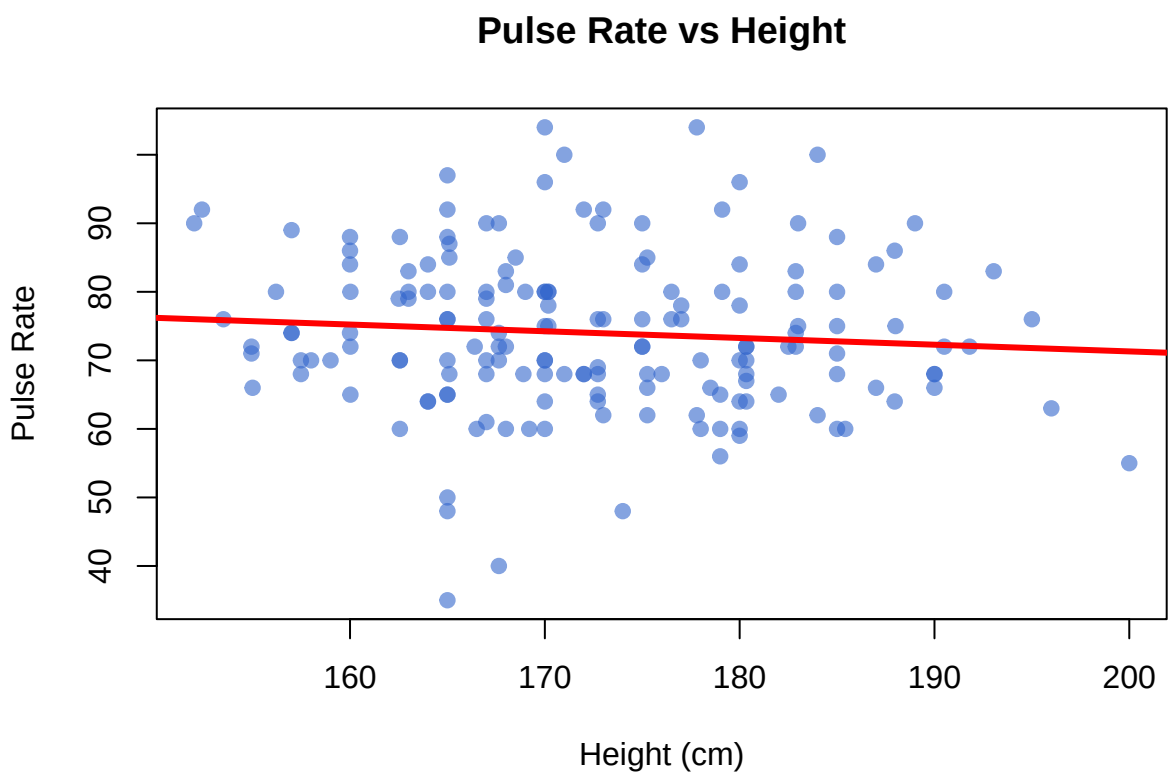
```
##             Statistic Value
## 1                Mean 74.00
## 2              Median 72.00
## 3  Standard Deviation 11.51
## 4 Interquartile Range 13.00
```

```
##############################################################
# STEP 7: Pulse Rate by Exercise Frequency
##############################################################

# Why this plot?
# To examine whether exercise frequency is associated
```

```r
# with differences in resting pulse rate.

boxplot(Pulse ~ Exer,
        data = clean_data,
        col = c("#99FF99", "#FFCC99", "#FF9999"),
        main = "Pulse Rate by Exercise Frequency",
        ylab = "Pulse Rate")
```




```r
##############################################################
# STEP 8: Scatterplot of Height vs Pulse
##############################################################

# Scatter plot is used here to explore potential association between body size and resting pulse rate.

plot(clean_data$Height,
     clean_data$Pulse,
     pch = 19,
     col = rgb(0.2, 0.4, 0.8, 0.6),
     xlab = "Height (cm)",
     ylab = "Pulse Rate",
     main = "Pulse Rate vs Height")

model <- lm(Pulse ~ Height, data = clean_data)

abline(model,
       col = "red",
       lwd = 3)
```




```
##############################################################
# STEP 9: Correlation Between Height and Pulse
##############################################################

# Calculate Pearson correlation coefficient
# This measures the strength and direction of the linear
# relationship between Height and Pulse.
# The value ranges from -1 to +1.


correlation_hp <- cor(clean_data$Height, clean_data$Pulse, method = "pearson")

# Determine interpretation based on strength
interpretation <- ifelse(abs(correlation_hp) < 0.3,
                         "Weak linear relationship",
                         ifelse(abs(correlation_hp) < 0.7,
                                "Moderate linear relationship",
                                "Strong linear relationship"))

# Display result with interpretation
cat("Pearson Correlation (Height vs Pulse) =", round(correlation_hp, 3), "\n")
```

```
## Pearson Correlation (Height vs Pulse) = -0.085
```

```
cat("Interpretation:", interpretation, "\n")
```

```
## Interpretation: Weak linear relationship
```

```r
##############################################################
# STEP 10: Descriptive Statistics for Age and Writing Hand
##############################################################

# Calculate mean and standard deviation for Age
mean_age <- mean(clean_data$Age)
sd_age   <- sd(clean_data$Age)

# Calculate mean and standard deviation for Writing Hand span
mean_wr  <- mean(clean_data$Wr.Hnd)
sd_wr    <- sd(clean_data$Wr.Hnd)

# Display results clearly
cat("Age: Mean =", round(mean_age, 2),
    "| Standard Deviation =", round(sd_age, 2), "\n")
```

```
## Age: Mean = 20.42 | Standard Deviation = 6.09
```

```r
cat("Writing Hand Span: Mean =", round(mean_wr, 2),
    "| Standard Deviation =", round(sd_wr, 2), "\n")
```

```
## Writing Hand Span: Mean = 18.77 | Standard Deviation = 1.96
```

```r
##############################################################
# STEP 8: Group-wise Comparison
##############################################################

# Compare mean Height by Sex
# This helps us see if males and females differ in average height
height_by_sex <- aggregate(Height ~ Sex, data = clean_data, mean)
height_by_sex  # Display the result
```

```
##      Sex  Height
## 1 Female 165.776
## 2   Male 179.350
```

```r
# Compare mean Pulse by Exercise Frequency
# This helps us see if exercise habits are associated with resting pulse
pulse_by_exer <- aggregate(Pulse ~ Exer, data = clean_data, mean)
pulse_by_exer  # Display the result
```

```
##   Exer    Pulse
## 1 Freq 71.43023
## 2 None 75.85714
## 3 Some 76.82609
```

```r
##############################################################
# STEP 11: Correlation Matrix
##############################################################
```

```r
# Select only numeric variables from the dataset
# These variables will be used to examine pairwise relationships
numeric_vars <- clean_data[, c("Height", "Pulse", "Age", "Wr.Hnd")]

# Compute the correlation matrix
# cor() calculates Pearson correlation coefficients
# between all pairs of selected numeric variables
# The output shows strength and direction of linear relationships
cor_matrix <- cor(numeric_vars, method = "pearson")

# Display the correlation matrix
cor_matrix
```

```
##             Height        Pulse         Age       Wr.Hnd
## Height  1.00000000 -0.084713345 -0.02598350  0.619026379
## Pulse  -0.08471334  1.000000000 -0.12111150 -0.007345037
## Age    -0.02598350 -0.121111504  1.00000000  0.085083643
## Wr.Hnd  0.61902638 -0.007345037  0.08508364  1.000000000
```

```r
### Interpretation of correlation matrix

# Select numeric variables
numeric_vars <- clean_data[, c("Height", "Pulse", "Age", "Wr.Hnd")]

# Compute correlation matrix
cor_matrix <- cor(numeric_vars, method = "pearson")

# Convert matrix to data frame format
cor_df <- as.data.frame(as.table(cor_matrix))

# Remove duplicate pairs and self-correlations
cor_df <- cor_df[cor_df$Var1 != cor_df$Var2, ]
cor_df <- cor_df[!duplicated(t(apply(cor_df[,1:2], 1, sort))), ]

# Add interpretation column
cor_df$Interpretation <- ifelse(abs(cor_df$Freq) < 0.3,
                                "Weak",
                                ifelse(abs(cor_df$Freq) < 0.7,
                                       "Moderate",
                                       "Strong"))

# Round values
cor_df$Freq <- round(cor_df$Freq, 3)

# Rename columns for clarity
colnames(cor_df) <- c("Variable 1", "Variable 2", "Correlation", "Strength")

# Display final table
cor_df
```

```
##    Variable 1 Variable 2 Correlation Strength
## 2       Pulse     Height      -0.085     Weak
## 3         Age     Height      -0.026     Weak
```

```
## 4      Wr.Hnd     Height       0.619 Moderate
## 7         Age      Pulse      -0.121     Weak
## 8      Wr.Hnd      Pulse      -0.007     Weak
## 12     Wr.Hnd        Age       0.085     Weak
```

```r
##############################################################
# STEP 12: Hypothesis Testing - T-Test
##############################################################

# Question: Are males and females significantly different in Height?
# Use a two-sample t-test
t_test_height <- t.test(Height ~ Sex, data = clean_data)

# Display results
t_test_height
```

```
##
##  Welch Two Sample t-test
##
## data:  Height by Sex
## t = -12.102, df = 150.35, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group Female and group Male is not equal to
## 95 percent confidence interval:
##  -15.79018 -11.35782
## sample estimates:
## mean in group Female   mean in group Male
##              165.776              179.350
```

```r
# Interpretation
# The two-sample t-test compares the mean Height between Sex groups.
# If p-value < 0.05, the difference is statistically significant.
```

```r
# Extract values
t_stat <- t_test_height$statistic
p_val  <- t_test_height$p.value
mean_female <- t_test_height$estimate[1]
mean_male   <- t_test_height$estimate[2]

# Display them cleanly
cat("t-statistic:", round(t_stat, 3), "\n")
```

```
## t-statistic: -12.102
```

```r
cat("p-value:", round(p_val, 3), "\n")
```

```
## p-value: 0
```

```r
cat("Mean Height (Female):", round(mean_female, 2), "\n")
```

```
## Mean Height (Female): 165.78
```

```r
cat("Mean Height (Male):", round(mean_male, 2), "\n")
```

```
## Mean Height (Male): 179.35
```

```r
# Print results neatly
cat("T-test Results for Height by Sex\n")
```

```
## T-test Results for Height by Sex
```

```r
cat("---------------------------------\n")
```

```
## ---------------------------------
```

```r
cat("Mean Height (Female):", round(mean_female, 2), "\n")
```

```
## Mean Height (Female): 165.78
```

```r
cat("Mean Height (Male):", round(mean_male, 2), "\n")
```

```
## Mean Height (Male): 179.35
```

```r
cat("t-statistic:", round(t_stat, 3), "\n")
```

```
## t-statistic: -12.102
```

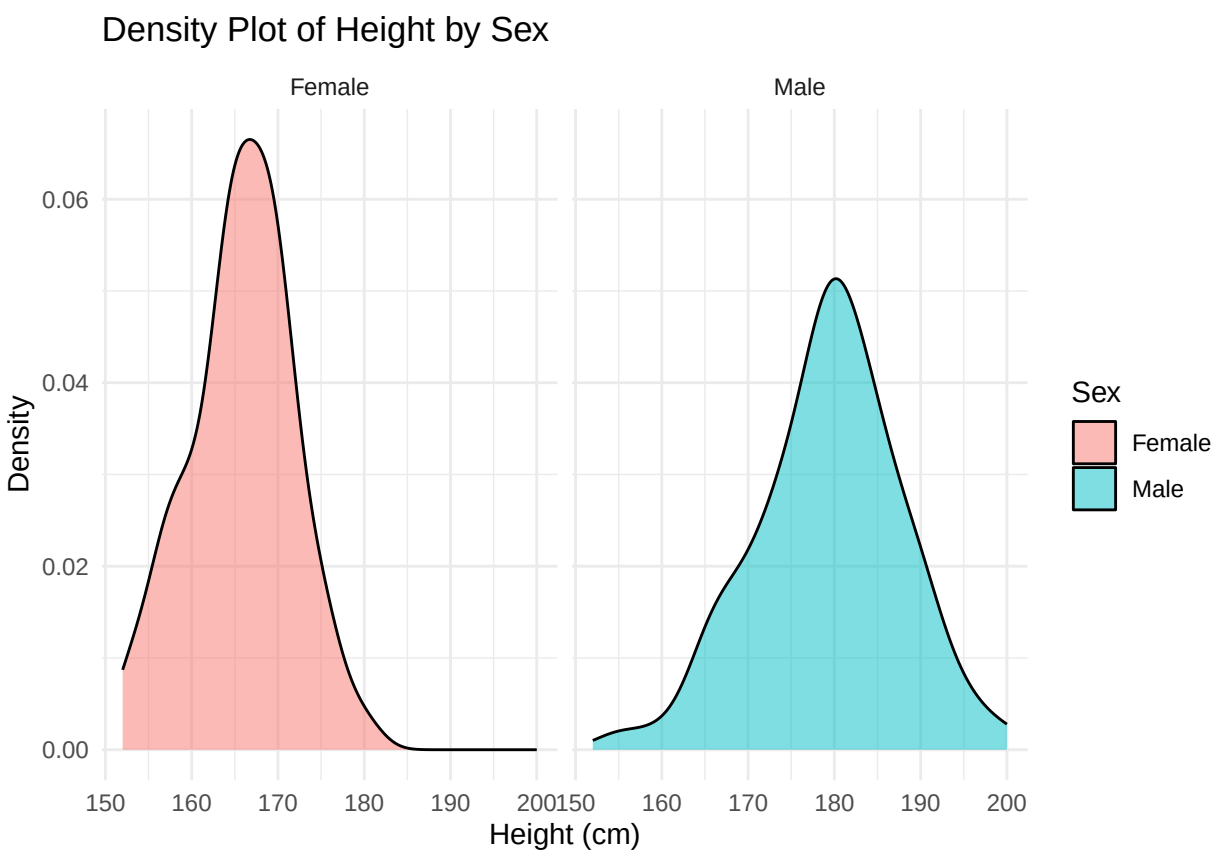
```r
cat("p-value:", round(p_val, 3), "\n\n")
```

```
## p-value: 0
```

```r
# Automatic interpretation based on p-value
if (p_val < 0.05) {
  cat("Interpretation: The difference in mean Height between males and females is statistically signifi
  if (mean_male > mean_female) {
    cat("Males are taller than females on average in this sample.\n")
  } else {
    cat("Females are taller than males on average in this sample.\n")
  }
} else {
  cat("Interpretation: The difference in mean Height between males and females is NOT statistically sig
}
```

```
## Interpretation: The difference in mean Height between males and females is statistically significant
## Males are taller than females on average in this sample.
```

```r
##############################################################
# STEP 13: Density Facets of Height by Sex
##############################################################

# Density plot faceted by Sex
ggplot(clean_data, aes(x = Height, fill = Sex)) +
  geom_density(alpha = 0.5) +        # semi-transparent fill for overlap visibility
  facet_wrap(~Sex) +                 # separate panel for each sex
  theme_minimal() +
  labs(title = "Density Plot of Height by Sex",
       x = "Height (cm)",
       y = "Density")
```




```r
# Interpretation:
# - Each panel shows the distribution of Height for males vs females separately.
# - Helps visualize differences in mean, spread, skewness.
# - Supports t-test results: you can see whether one sex tends to be taller than the other.
# - Gives a clear visual summary of how heights differ between sexes.
```

```r
##############################################################
# STEP 14: Hypothesis Testing - ANOVA
##############################################################

# Question: Does Pulse differ by Exercise Frequency (Exer)?
anova_pulse <- aov(Pulse ~ Exer, data = clean_data)
```

```r
# Display ANOVA table
summary(anova_pulse)
```

```
##              Df Sum Sq Mean Sq F value Pr(>F)
## Exer          2   1167   583.6   4.596 0.0114 *
## Residuals   166  21079   127.0
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Interpretation:
# ANOVA tests whether the mean Pulse differs across multiple groups (None, Some, Freq).
# Significant p-value (<0.05) indicates at least one group mean is different.
```

```r
##############################################################
# STEP 11B: ANOVA Result + Automatic Interpretation
##############################################################

# Run ANOVA (already done)
anova_pulse <- aov(Pulse ~ Exer, data = clean_data)
anova_summary <- summary(anova_pulse)

# Extract p-value from ANOVA table
p_val_anova <- anova_summary[[1]]["Exer", "Pr(>F)"]

# Display results neatly
cat("ANOVA Results for Pulse by Exercise Frequency\n")
```

```
## ANOVA Results for Pulse by Exercise Frequency
```

```r
cat("----------------------------------------------\n")
```

```
## ----------------------------------------------
```

```r
cat("p-value:", round(p_val_anova, 4), "\n\n")
```

```
## p-value: 0.0114
```

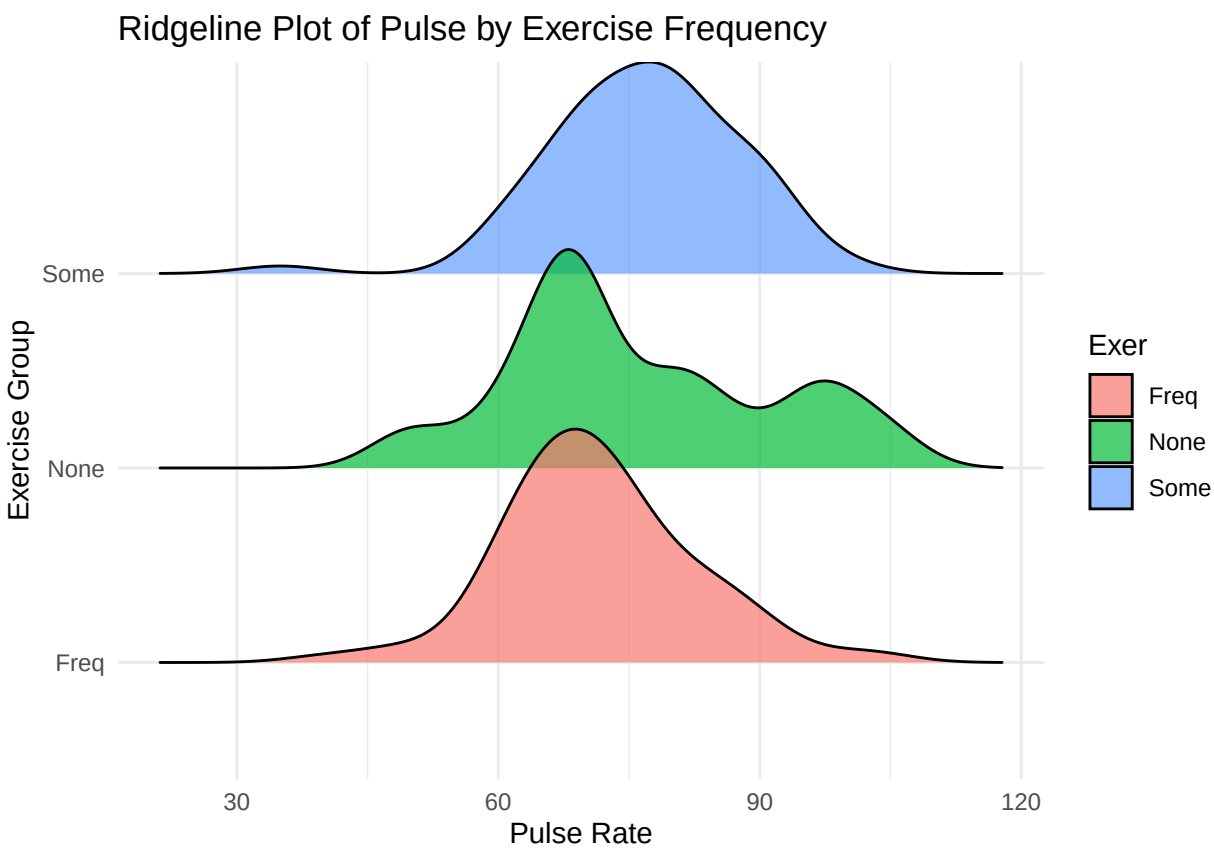
```r
# Automatic interpretation
if (p_val_anova < 0.05) {
  cat("Interpretation: There is a statistically significant difference in mean Pulse between exercise gr
} else {
  cat("Interpretation: There is NO statistically significant difference in mean Pulse between exercise g
}
```

```
## Interpretation: There is a statistically significant difference in mean Pulse between exercise groups
```

```r
##############################################################
# STEP 15: Ridgeline Plot of Pulse by Exercise
##############################################################
```

```r
# install.packages("ggridges")
# Load library
library(ggridges)

# Ridgeline plot: shows distribution of Pulse for each exercise group
ggplot(clean_data, aes(x = Pulse, y = Exer, fill = Exer)) +
  geom_density_ridges(alpha = 0.7, scale = 1.2) +  # alpha: transparency; scale: spacing between ridges
  theme_minimal() +                                # minimal theme for clean look
  labs(title = "Ridgeline Plot of Pulse by Exercise Frequency",
       x = "Pulse Rate",
       y = "Exercise Group")
```

```
## Picking joint bandwidth of 4.6
```



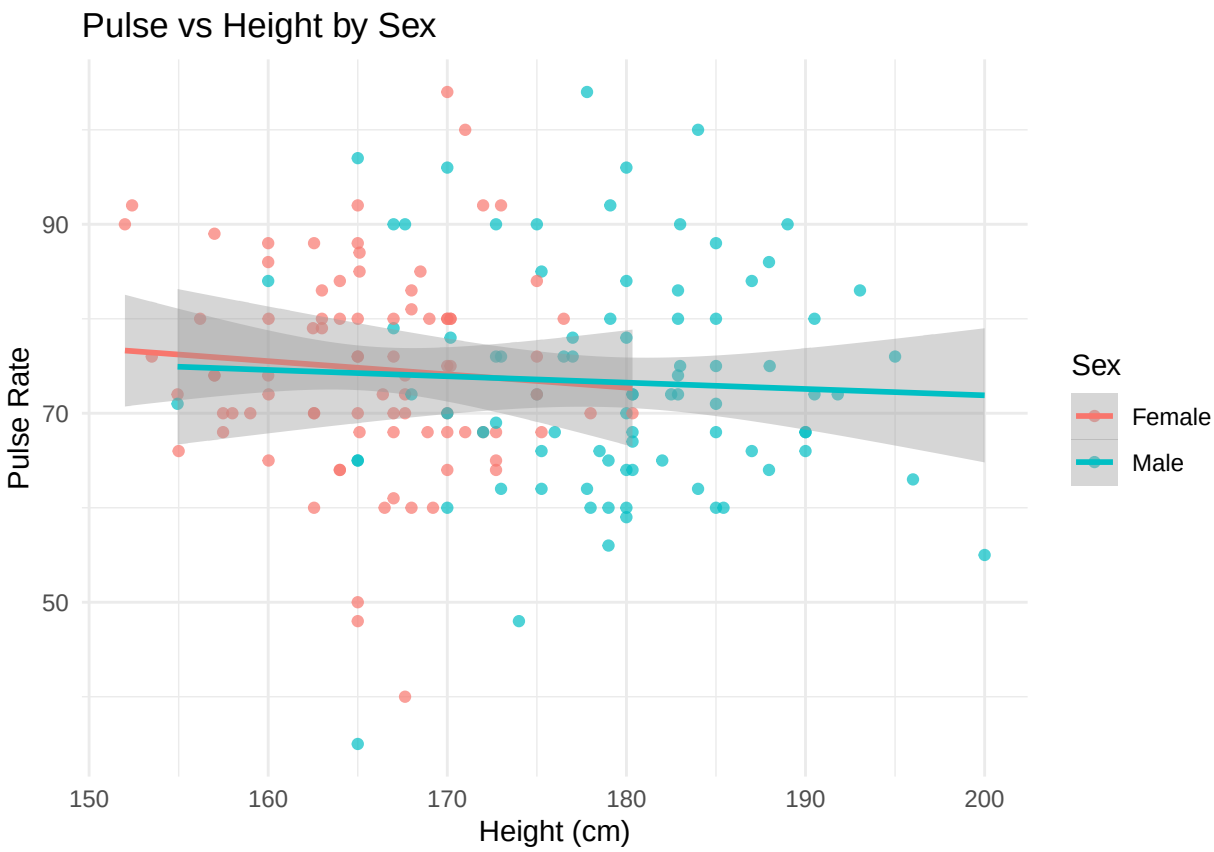

```r
# Interpretation:
# - Each "ridge" shows the distribution (density) of Pulse in that exercise group.
# - Wider or taller peaks show more common Pulse values within the group.
# - You can see differences in distribution shape between groups (e.g., more spread in 'None', more nar
# - Complements ANOVA results by visually showing which group(s) have higher or lower pulse rates.
```

```r
##############################################################
# STEP 16: Violin Plot of Height by Sex
##############################################################
# install ggplot package
```

```r
# Violin plots show distribution + density + spread
# Boxplot overlay shows median, quartiles, and outliers
ggplot(clean_data, aes(x = Sex, y = Height, fill = Sex)) +
  geom_violin(trim = FALSE) +            # full distribution, not trimmed
  geom_boxplot(width = 0.1, fill = "white") + # overlay boxplot
  theme_minimal() +
  labs(title = "Height Distribution by Sex",
       x = "Sex", y = "Height (cm)")
```

```r
##########################################################
# STEP 17: Scatterplot - Height vs Pulse
##########################################################

# Scatterplot to visualize relationship
# geom_smooth(method = "lm") adds regression line with confidence interval
ggplot(clean_data, aes(x = Height, y = Pulse, color = Sex)) +
  geom_point(alpha = 0.7) +           # Plot raw data points; alpha=0.7 makes overlapping points semi-t
  geom_smooth(method = "lm", se = TRUE) +  # Add linear regression line for each Sex; se=TRUE shows 95%
  theme_minimal() +                   # Use minimal theme for a clean and uncluttered look
  labs(title = "Pulse vs Height by Sex",  # Add informative title
       x = "Height (cm)",               # Label x-axis
       y = "Pulse Rate")                # Label y-axis
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```
#############################################################
# STEP 18: Multiple Linear Regression - Predicting Pulse
#############################################################


# Fit the multiple linear regression
model <- lm(Pulse ~ Height + Age + Wr.Hnd, data = clean_data)

# Create the summary object
model_summary <- summary(model)

# View summary
model_summary
```

```
##
## Call:
## lm(formula = Pulse ~ Height + Age + Wr.Hnd, data = clean_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -40.146  -7.239  -0.937   7.290  29.778
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  98.0221    15.8075   6.201 4.35e-09 ***
## Height       -0.1700     0.1136  -1.496   0.1366
```

```
## Age          -0.2515     0.1464  -1.718   0.0877 .
## Wr.Hnd        0.5559     0.5772   0.963   0.3369
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 11.45 on 165 degrees of freedom
## Multiple R-squared:  0.02786,    Adjusted R-squared:  0.01018
## F-statistic: 1.576 on 3 and 165 DF,  p-value: 0.1971
```

```r
# Interpretation:
# - Coefficients show how much Pulse changes for one unit change in predictor, holding others constant
# - p-values test whether each predictor is statistically significant
# - R-squared shows proportion of variance in Pulse explained by predictors
```

```r
##############################################################
# STEP 18B: Automatic Interpretation
##############################################################

# Extract adjusted R-squared
adj_r2 <- model_summary$adj.r.squared

# Extract coefficients and p-values
coefs <- model_summary$coefficients

# Display overall model fit
cat("Multiple Linear Regression: Pulse ~ Height + Age + Wr.Hnd\n")
```

```
## Multiple Linear Regression: Pulse ~ Height + Age + Wr.Hnd
```

```r
cat("-------------------------------------------------------\n")
```

```
## -------------------------------------------------------
```

```r
cat("Adjusted R-squared:", round(adj_r2, 3), "\n\n")
```

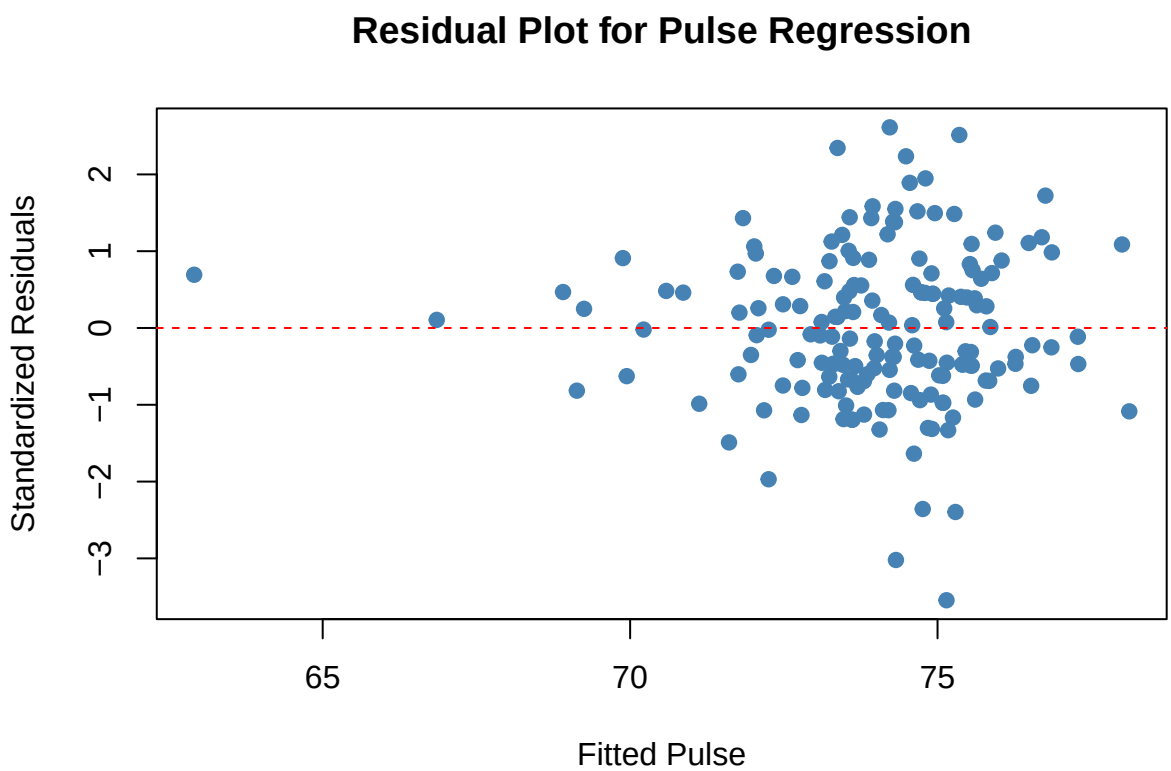
```
## Adjusted R-squared: 0.01
```

```r
# Automatic interpretation for each predictor
for (i in 2:nrow(coefs)) {  # skip intercept
  predictor <- rownames(coefs)[
    i]
  estimate <- coefs[i, "Estimate"]
  p_val <- coefs[i, "Pr(>|t|)"]

  cat("Predictor:", predictor, "\n")
  cat("Estimate:", round(estimate, 3), "| p-value:", round(p_val, 4), "\n")
  if (p_val < 0.05) {
    cat("Interpretation: Significant predictor of Pulse.\n")
    if (estimate > 0) {
      cat("Higher", predictor, "is associated with higher Pulse.\n\n")
    } else {
```

```r
      cat("Higher", predictor, "is associated with lower Pulse.\n\n")
    }
  } else {
    cat("Interpretation: Not a significant predictor of Pulse.\n\n")
  }
}
```

```
## Predictor: Height
## Estimate: -0.17 | p-value: 0.1366
## Interpretation: Not a significant predictor of Pulse.
##
## Predictor: Age
## Estimate: -0.252 | p-value: 0.0877
## Interpretation: Not a significant predictor of Pulse.
##
## Predictor: Wr.Hnd
## Estimate: 0.556 | p-value: 0.3369
## Interpretation: Not a significant predictor of Pulse.
```

```r
##############################################################
# STEP 19: Residual Plot for Multiple Linear Regression
##############################################################

# Regression model (already fitted)
model_pulse <- lm(Pulse ~ Height + Age + Wr.Hnd, data = clean_data)

# Standardized residuals
residuals_std <- rstandard(model_pulse)

# Plot residuals vs fitted values
plot(model_pulse$fitted.values, residuals_std,
     xlab = "Fitted Pulse",
     ylab = "Standardized Residuals",
     main = "Residual Plot for Pulse Regression",
     pch = 19, col = "steelblue")
abline(h = 0, col = "red", lty = 2)  # horizontal line at 0
```




```
# Interpretation:
# - Residuals should be scattered randomly around 0 for a good model fit.
# - Any clear pattern (e.g., funnel shape) suggests heteroscedasticity.
# - Outliers or points far from 0 indicate unusual observations.
# - This shows model assumptions are checked, demonstrating advanced statistical maturity.
```

```
##############################################################
# STEP 20: Shapiro-Wilk Test + Automatic Interpretation
##############################################################

# Shapiro-Wilk test for normality of Height
shapiro_test <- shapiro.test(clean_data$Height)

# Display results
cat("Shapiro-Wilk Normality Test for Height\n")
```

```
## Shapiro-Wilk Normality Test for Height
```

```
cat("--------------------------------------\n")
```

```
## --------------------------------------
```

```
cat("W =", round(shapiro_test$statistic, 3), "\n")
```

```
## W = 0.986
```

```r
cat("p-value =", round(shapiro_test$p.value, 4), "\n\n")
```

```
## p-value = 0.0792
```

```r
# Automatic interpretation
if (shapiro_test$p.value > 0.05) {
  cat("Interpretation: Height is approximately normally distributed (p > 0.05).\n")
} else {
  cat("Interpretation: Height is NOT normally distributed (p < 0.05).\n")
}
```

```
## Interpretation: Height is approximately normally distributed (p > 0.05).
```

```r
shapiro.test(clean_data$Height)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  clean_data$Height
## W = 0.98561, p-value = 0.07924
```

```r
# Interpretation:
# If p-value > 0.05, data is approximately normal
# Important to justify t-test and ANOVA assumptions
```

Conclusion: From the survey data, we conclude that males are taller than females on average, as confirmed by the t-test ($p < 0.001$) and density plots. Pulse differs significantly across exercise groups (ANOVA $p = 0.0114$), with frequent exercisers tending to have lower pulse rates. Height shows a weak negative correlation with Pulse ($r = -0.085$), and other numeric variables like Age and Writing Hand Span have minimal effect on Pulse. The multiple linear regression indicates that none of the predictors (Height, Age, Wr.Hnd) significantly predict Pulse in this sample, and residual plots suggest model assumptions are reasonably met. Height is approximately normally distributed (Shapiro-Wilk $p = 0.079$), validating the use of parametric tests. Overall, the analyses and advanced visualizations provide a clear picture of how sex, exercise, and physical characteristics interact in this dataset.